

Condensate Systems: Compliance, SOP & Optimization Guide

Purpose:

Recovers hot condensate from steam systems to conserve energy, reduce water waste, and protect equipment.

Why It's Ignored:

Systems are often hidden or overlooked until water hammer, flooding, or pump failure occurs.

Must-Have Docs:

- Trap survey logs
- Flash tank maintenance records
- Return line insulation inspection
- SOPs for blowdown and drain handling

Compliance Officials:

ASME Section I & VIII, OSHA 1910.169, State Boiler Inspectors

PPE Required:

Thermal gloves, face shield, safety glasses, hearing protection (for trap testing)

Installation Practices:

- Sloped piping for drainage
- Properly vented flash tanks
- Insulated lines
- Mechanical vs. electric pump specs documented

Must Checks:

- Steam trap performance (ultrasonic or temp test)
- Condensate return pump runtime and failure rate
- Flash tank pressure and flow - Evidence of water hammer or pipe corrosion

Regulations & Codes:

ASME, OSHA 1910.169, Local mechanical/plumbing code